Sandhya Jayaraman

sjayaraman@ucsd.edu Sandhya's LinkedIn Personal Website

Education

2022 – present **Ph.D., UC San Diego** Computer Science and Engineering, *Advanced to Candidacy.*
Advisor: Dr. Laurel Riek
My research focuses on a care ethics based approach to the design of robots to support clinicians and patients in acute care, and the application of formal methods towards a safe, and privacy-aware implementation of such robots.

2021 – 2022 **MS., UC San Diego** Computer Science and Engineering.

2016 – 2020 **BE., Visveswaraya Technological University** Computer Science and Engineering.

Research Experience

2021 - present **Graduate Student Researcher** Healthcare Robotics Lab advised by **Dr. Laurel Riek**.

Disability-forward Design of Robots for Acute Care

- Led a disability-forward, research through design project with people with disabilities and / or chronic illness (PwD/CI) to explore the design of robots that might help PwD/CI's access to and experience of acute care.
- Identified novel robot roles PwD/CI envisioned, and surfaced key values and priorities for robots to support PwD/CI's access and experience of acute care.

Robots for Acute Cancer Care

- Led co-design research with emergency medicine clinicians and people with cancer to design robots to support acute cancer care
- Generated broad provocations rooted in care ethics and activity theory to inform the design of future cancer support robots.

Clinically Assistive Robots for Acute Care

- Led collaborations with emergency medicine clinicians to co-design a clinically assistive robot (CAR) system to support clinicians.
- Leveraged formal methods to implement a correct-by-construction CAR, and conducted evaluations in two real-world emergency departments.
- Identified broader socioethical implications, and future research directions for CARs.

Privacy and Utility Perceptions of Social Robots

- Designed and conducted a large scale ($N = 239$) online quantitative study to explore people's privacy perceptions of social robots.
- Worked with a professional animator to develop animated design fiction videos of high cognitive fidelity as prompts for online participants.
- Identified key factors (e.g., context of robot operation) influencing privacy perceptions.

Apr - Jun 2021 **Research Student** Programming Languages and Systems Security Lab advised by **Dr. Deian Stefan**

Jan - Oct 2020 **Research Intern** Indian Institute of Science (IISc) Supercomputer Education and Research Center (SERC) advised by **Dr. R Govindarajan**

May - Jul 2018 **Research Intern** Indian Institute of Science Education and Research (IISER) Quantum Computing Lab advised by **Dr. Prasanta Panigrahi**

May - Jul 2017 **Research Intern** IISER Mathematics Lab advised by **Dr. Amit Kulshrestha**

Publications

1. **S. Jayaraman**, P. Ghosh, S. Kim, S. Satyadharma, A. Taylor, C. Coyne, and L. D. Riek, "Designing care-fully: Robots for acute cancer care," in *Proceedings of the 2026 ACM/IEEE international conference on human-robot interaction*, **Best paper honorable mention** [Acceptance Rate: 23%], 2026.
2. **S. Jayaraman**, A. Violette, U. L. Lou, S. Mani, D. Prakash, L. Oyama, C. Coyne, H. Kress-Gazit, and L. Riek, "Car-em: A synthesis-based clinically assistive robot system for emergency medicine," *ACM Transactions on Human-Robot Interaction*, 2026.
3. **S. Jayaraman**, D. Masanam, P. Ghosh, A. Kubota, and L. D. Riek, "Robots for care: Exploring downstream socio-ethical effects and upstream interventions," 2026.
4. **S. Jayaraman**, A. Violette, U. L. Lou, S. Mani, D. Prakash, L. C. Oyama, C. Coyne, H. Kress-Gazit, and L. D. Riek, "Contextualizing clinically assistive robots for emergency departments," *ISMR Workshop on Robotics for Nursing*, 2025.
5. **S. Jayaraman**, E. K. Phillips, D. Church, and L. D. Riek, "Privacy and utility perceptions of social robots in healthcare," *Computers in Human Behavior: Artificial Humans*, vol. 2, no. 1, p. 100 039, 2024.
6. **S. Jayaraman**, E. K. Phillips, D. Church, and L. D. Riek, "Social robots in healthcare: Characterizing privacy considerations," in *Companion of the 2024 ACM/IEEE International Conference on Human-Robot Interaction*, 2024, pp. 568–572.
7. **S. Jayaraman**, D. Cruz-Sandoval, A. Kubota, and L. D. Riek, "What a healthcare worker wants, what a disabled person needs: Exploring stakeholder design tensions in assistive robotics," in *HRI 2024 Workshop on Assistive Applications, Accessibility, and Disability Ethics (A3DE)*, 2024.
8. R. Jha, D. Das, A. Dash, **S. Jayaraman**, B. K. Behera, and P. K. Panigrahi, "A novel quantum n-queens solver algorithm and its simulation and application to satellite communication using ibm quantum experience," *arXiv preprint arXiv:1806.10221*, 2018.

Supervisees

Sep 2026 - present	Anika Dontu (BS Computer Science, UCSD)
Jul 2023 - present	U Lam Lou (MS Computer Science, UCSD).
Jun - Aug 2026	McKenzie Davis Watkins (BS Information Science, University of Maryland)
	Thane Douglass (BS Computer Science, Howard University)
Jan 2024 - Jun 2026	Srutilaya Mani (BS Mathematics / Computer Science, UCSD).
Jun - Aug 2024	Divya Prakash (BS Public Health / Computer Science, Austin College).
Jan - May 2023	Andrea Behar-yutkewich (BS Cognitive Science, UCSD).

Teaching Experience

Spring 2024	**TA, UCSD CSE 276B** Human Robot Interaction (Graduate level).
Winter, Spring 2022	**TA, UCSD CSE 20** Discrete Mathematics.
Fall 2021	**TA, UCSD CSE 202** Design and Analysis of Algorithms (Graduate level).
Summer 2021	**TA, UCSD CSE 105** Theory of Computing.
Spring 2021	**TA, UCSD CSE 240A** Computer Architecture (Graduate level).
2021 - 2022	**TA, UCSD CSE 599** Mentored first-time Teaching Assistants.
Winter 2021	**TA, UCSD CSE 101** Design and Analysis of Algorithms.

Academic Service

2025 **Reviewer.** ACM Transactions on Human Robot Interaction (THRI).

Academic Service (continued)

	Reviewer. ACM Transactions on Computing for Healthcare (ACM Health).
2021 - 2025	**Reviewer.** ACM/IEEE Conference on Human Robot Interaction (HRI).
2022	**Reviewer.** Communications of the ACM (CACM).

Awards and Outreach

March 2026	**Best Paper Honorable Mention** "Designing Care-Fully: Robots for Acute Cancer Care" at HRI'26.
	Awardee. IEEE RAS Travel Support Award for HRI'26.
	Organizer. HRI'26 Workshop on Robots for Care: Exploring downstream socio-ethical effects and upstream interventions.
May 2025	**Presenter.** International Symposium on Medical Robotics.
2024 - 2025	**Secretary.** RoboGrads (Student-run org. for graduate students in robotics), UCSD.
2024 - present	**Student Rep.** Diversity, Equity and Inclusion (DEI) Committee, UCSD CSE.
Sep 2024	**DEI Attendee.** Tapia Conference, Association for Computing Machinery.
Apr 2024	**Member.** Grad Cohort Workshop for Women, Computing Research Association.
Mar 2024	**Poster Presenter.** IEEE/ACM International Conference on Human-Robot Interaction.